Data Structures and Algorithms · 21CSC201J

Unit 2 · Session 8 · SLO 1 — Solved Worksheet

Name: DEEP SAND DEEP SAND | **Reg. No:** RA2511026010557

Unit â€“ 2 : Session â€“ 8 : SLO â€“ 1 | Circular Singly Linked List â€” Insertion

■ I. Insert a New Node VAL at the Beginning

• Step 1: IF AVAIL = NULL
• Write OVERFLOW
• Go to Step 11
• [END OF IF]
• Step 2: SET NEW_NODE = AVAIL
• Step 3: SET AVAIL = AVAIL -> NEXT
• Step 4: SET NEW_NODE -> DATA = VAL
• Step 5: SET PTR = START
• Step 6: Repeat Step 7 while PTR -> NEXT != START
• Step 7: SET PTR = PTR -> NEXT
• [END OF LOOP]
• Step 8: SET NEW_NODE -> NEXT = START
• Step 9: SET PTR -> NEXT = NEW_NODE
• Step 10: SET START = NEW_NODE
• Step 11: EXIT

■ II. Insert a New Node at the End

• Step 1: IF AVAIL = NULL
• Write OVERFLOW
• Go to Step 10
• [END OF IF]
• Step 2: SET NEW_NODE = AVAIL
• Step 3: SET AVAIL = AVAIL -> NEXT
• Step 4: SET NEW_NODE -> DATA = VAL
• Step 5: SET NEW_NODE -> NEXT = START
• Step 6: SET PTR = START
• Step 7: Repeat Step 8 while PTR -> NEXT != START
• Step 8: SET PTR = PTR -> NEXT
• [END OF LOOP]
• Step 9: SET PTR -> NEXT = NEW_NODE
• Step 10: EXIT